

Workload Management and Pricing

07

Topics

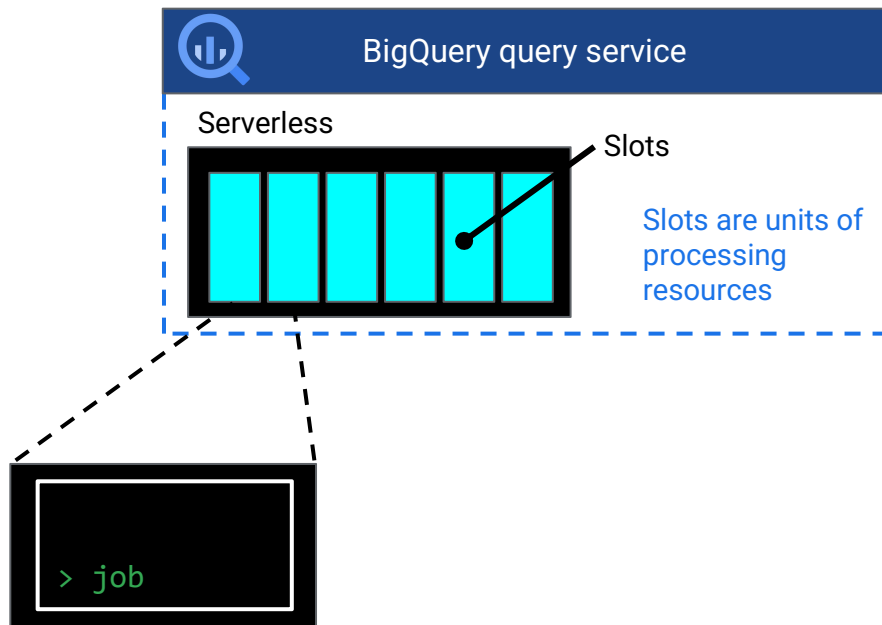
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|----|------------------------------|
| 01 | BigQuery Slots |
| 02 | Pricing Models and Estimates |
| 03 | Slot Reservations |
| 04 | Controlling Costs |





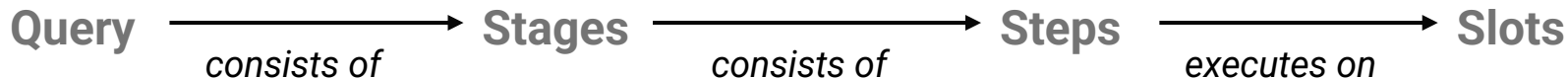
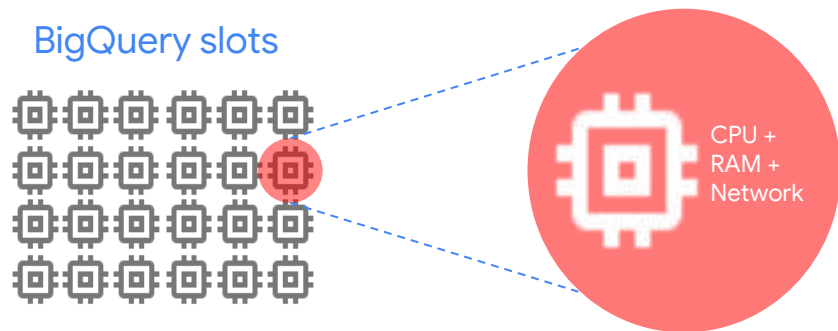
BigQuery Slots

Slots are units of resources that are consumed when a query runs



BigQuery slots execute your SQL queries

- A BigQuery slot is a unit of computational capacity (in terms of CPU, memory, and networking resources) required to execute workloads (SQL queries).
- A slot commitment is equivalent to buying a dedicated portion of a BigQuery cluster.
- BigQuery compute cluster is a regional resource (zonal redundancy)
- BigQuery automatically calculates the number of slots required by each query based on query size and complexity.



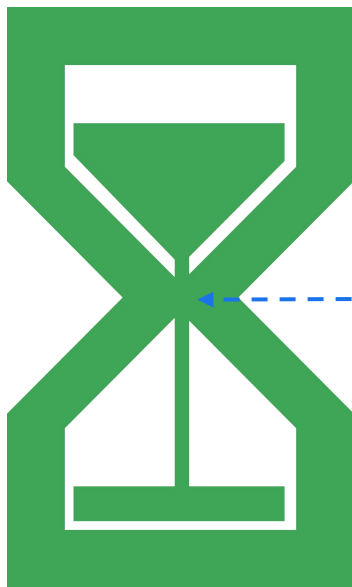
Estimating the right BigQuery slots allocation is critical

General guideline:

We recommend that you plan for 2,000 BigQuery slots for every 50 medium-complexity queries simultaneously.

50 queries = 2,000 slots

Quota: 100 concurrent queries, but this can be raised

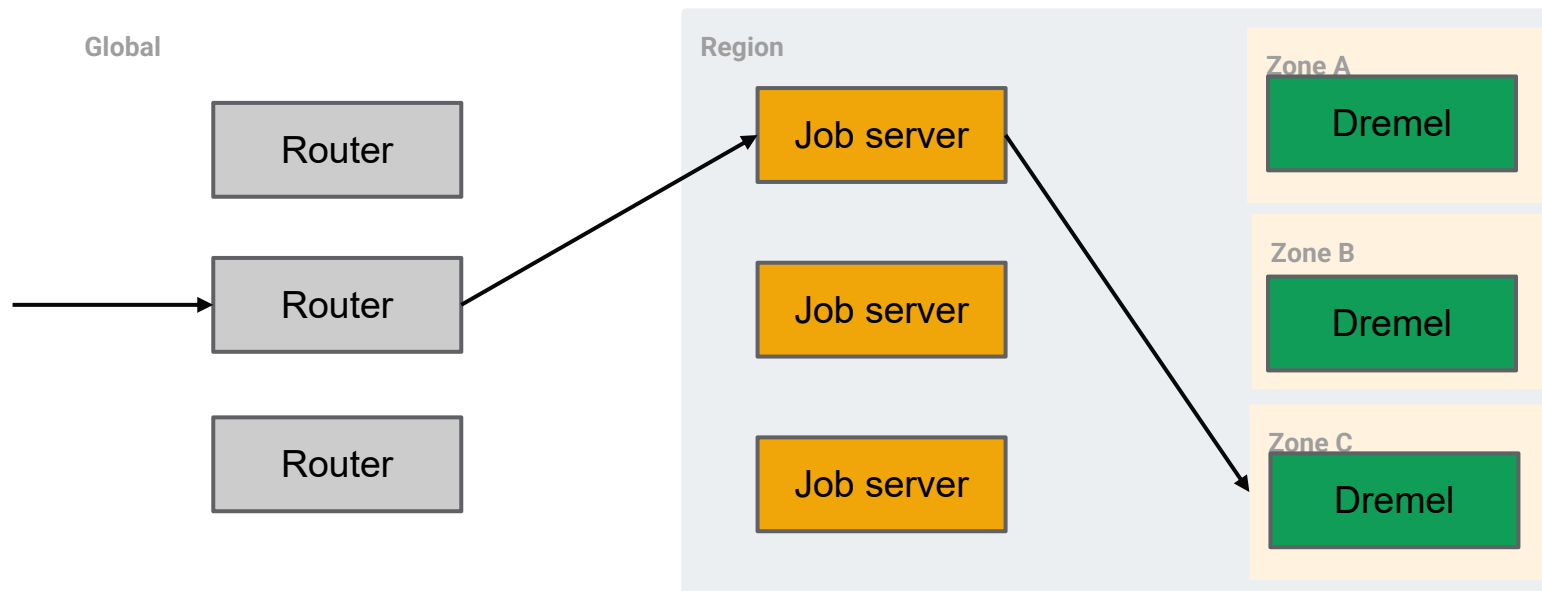


Slots available

More slots = Faster execution rate

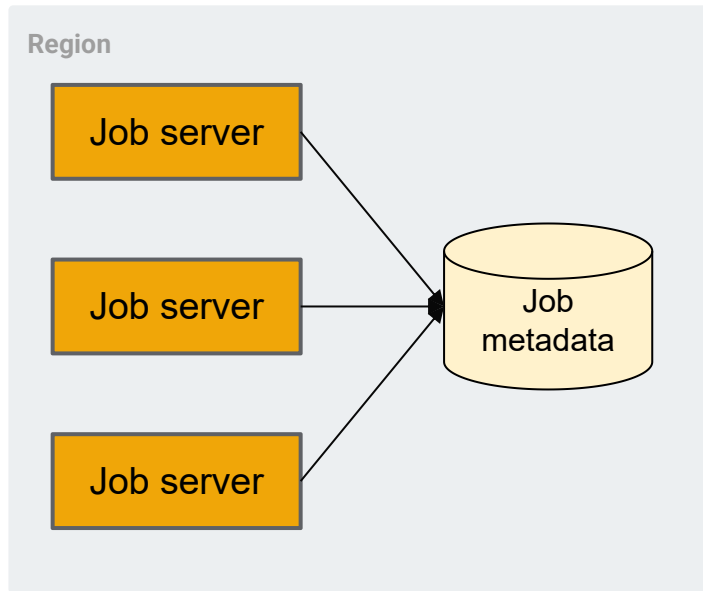
More concurrent queries = Slower execution rate

Query overview

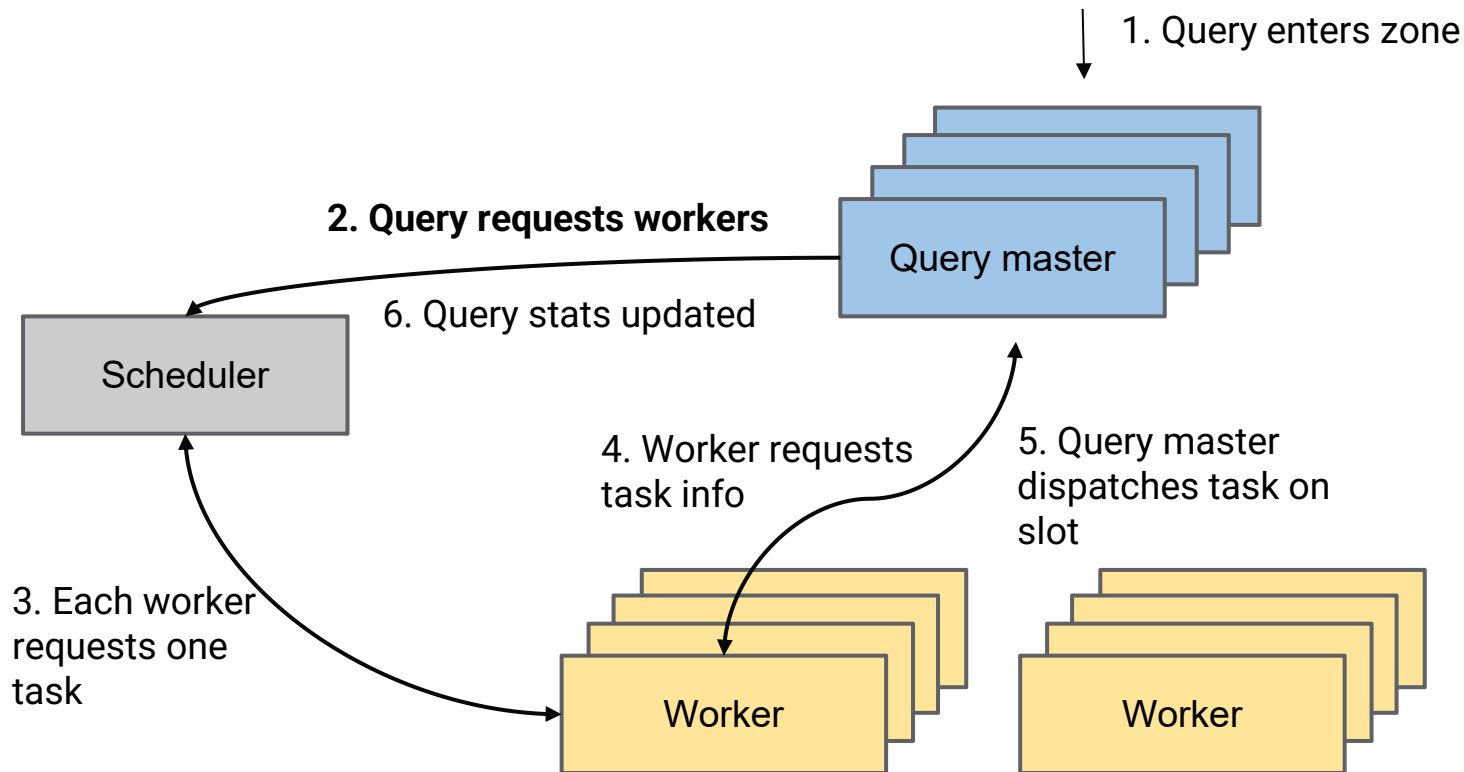


Query overview: Job queuing

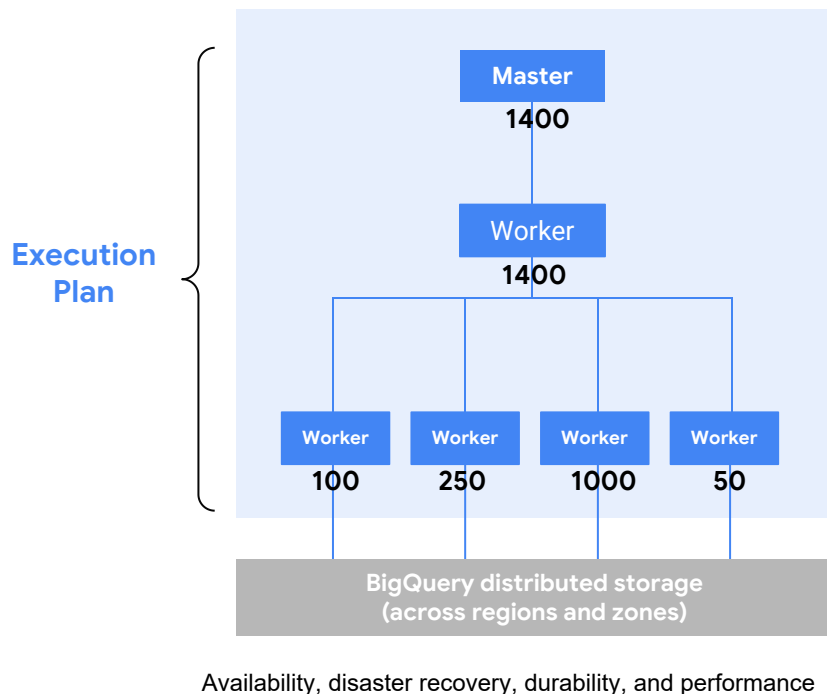
- Jobs start in the **PENDING** state.
 - Can transition to either **RUNNING** or **DONE** (due to timeout).
 - Most jobs immediately enter the **RUNNING** state.
- Jobs defer their **RUNNING** transition when:
 - **BATCH** priority: Always defer at least 1 minute; defer longer if awaiting quota or the individual server is nearing capacity.
 - **INTERACTIVE** priority: Never.
- The job server will periodically re-evaluate the deferment, with exponential backoff.



Query overview: Scheduling lifecycle



Query processing: Example



```
SELECT COUNT(*)  
FROM `bigquery-public-  
data.new_york_citibike.citibike_trips`  
WHERE start_station_name LIKE "%Broadway%"
```

← Stage 2: Sum (1 worker)

← Stage 1: Filter, count (4 workers)

Read data from source (bq storage or federated) (Network Capacity = \$)

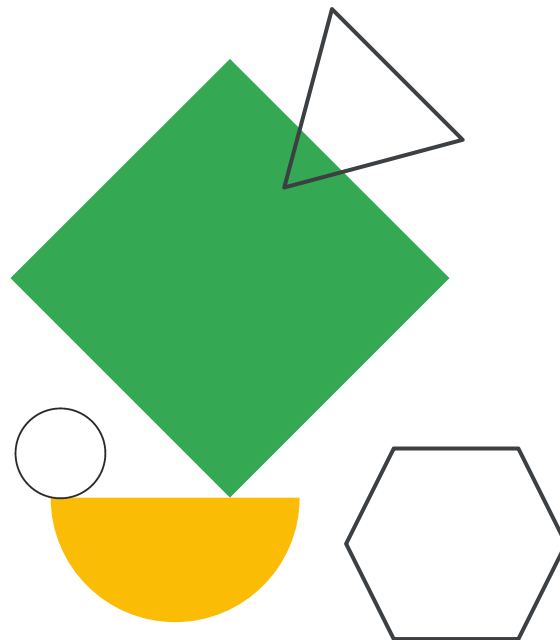
Operations \ Calculations (CPU Capacity = \$)

Shuffle & Store Intermediate results (Memory Capacity = \$)

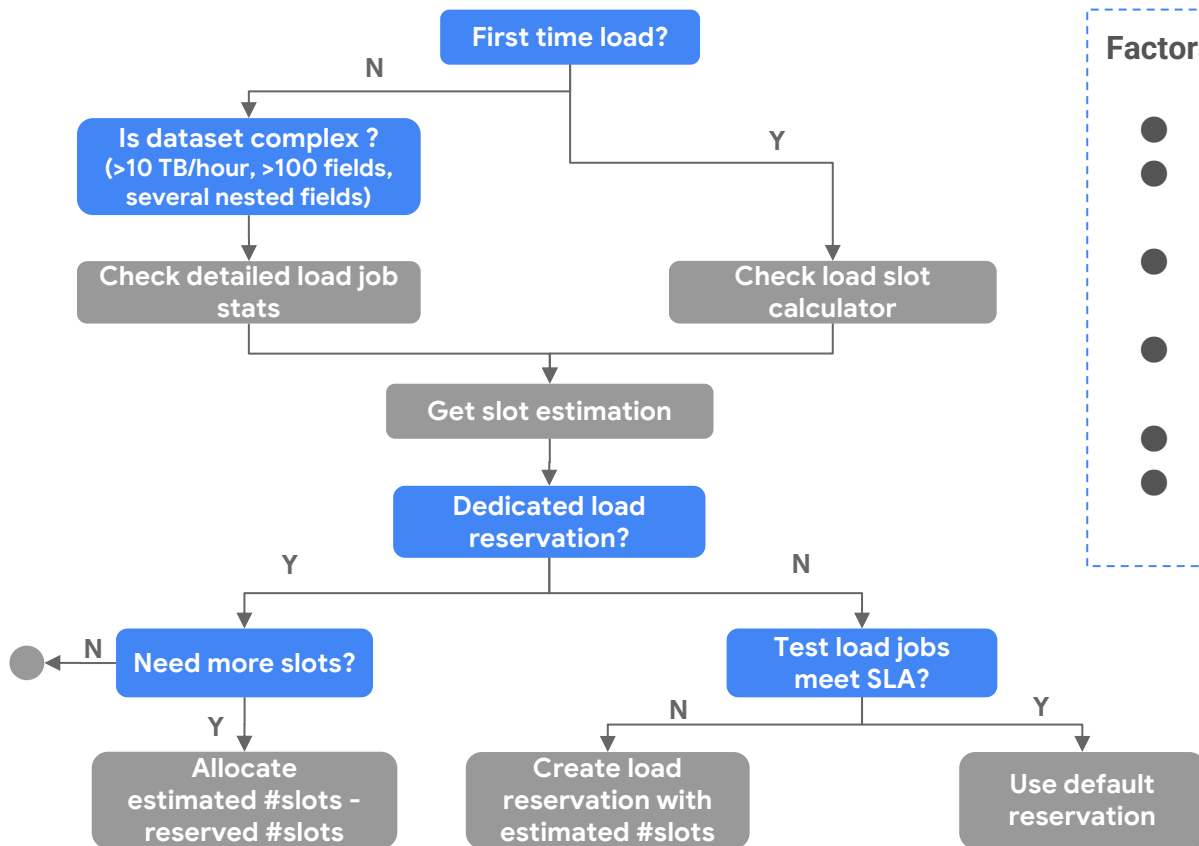
Demo

Determining slot usage for a query

Using SQL or the API



Slot projection: Loading data



Factors for load slot estimations:

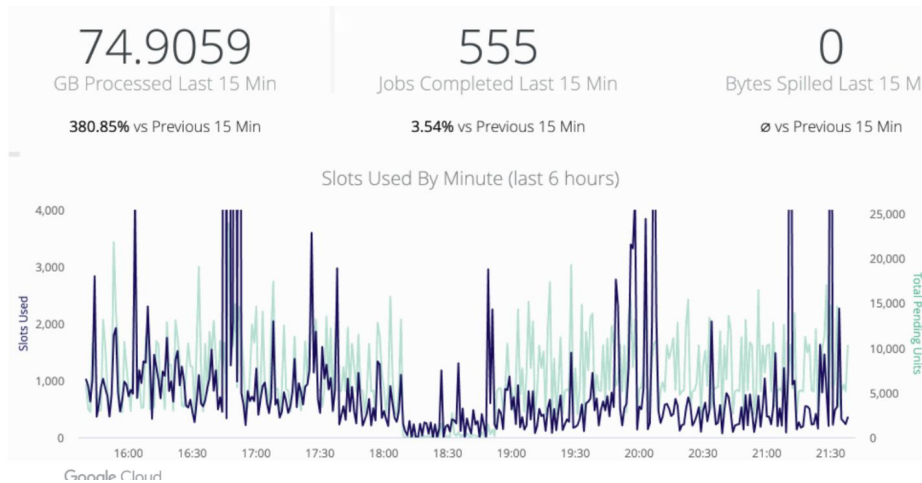
- Dataset size
- Dataset complexity: #fields, # nested/repeated fields
- Table schema: Is the table partitioned or clustered?
- Load frequency: Hourly, Daily, Every n-hours
- Load SLA
- Historical load throughput

[Load Slot Calculator](#)

Slot projection: Look for the signals

How do we know whether we need more slots..??

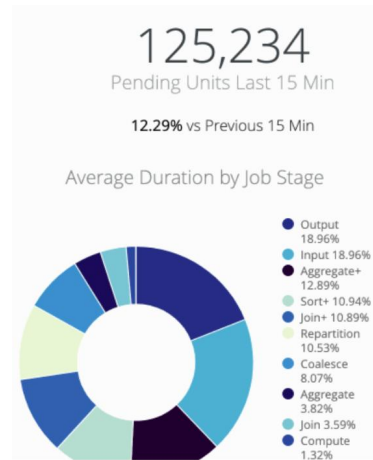
- Monitor trend (week over week, month over month)
 - Are more users being onboarded within the same slot allocation?
 - Monitor user growth/trend



Slot projection: Look for the signals

How do we know whether we need more slots..??

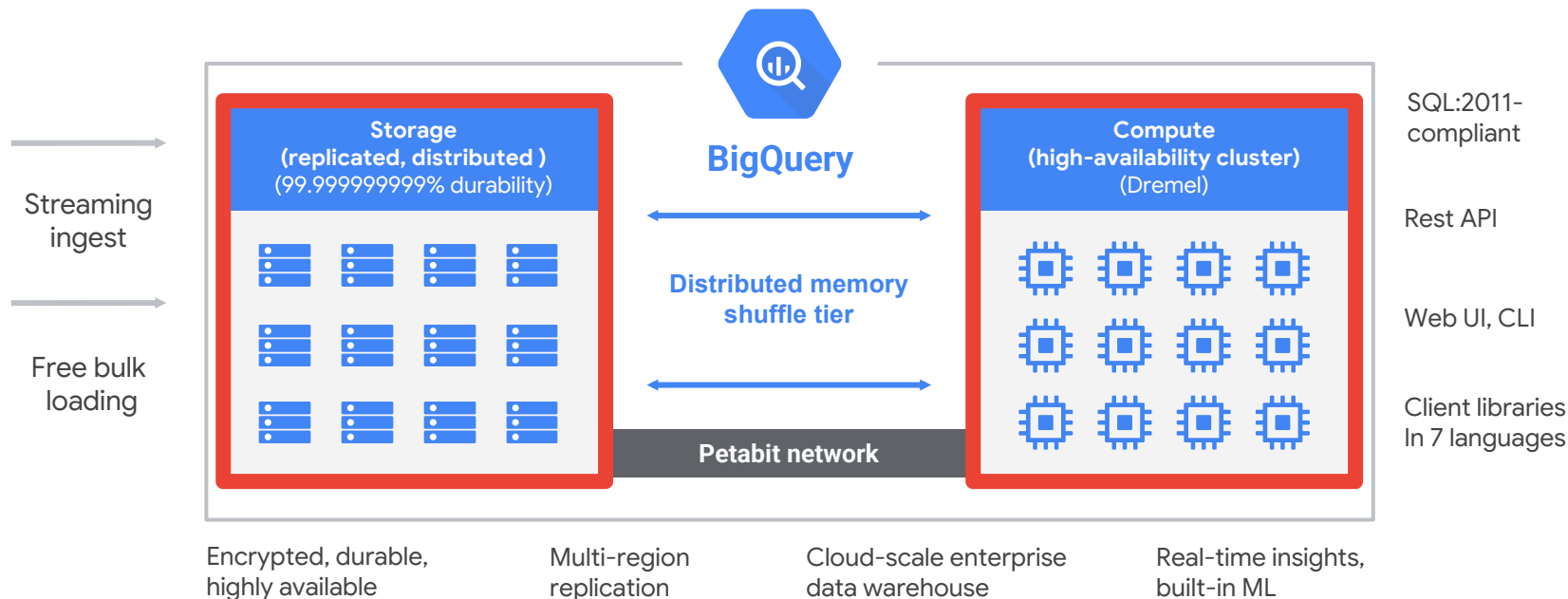
- More slots are needed for user queries if, for a given slot allocation, the following is observed:
 - Concurrency - consistently increasing
 - Throughput - consistently decreasing
 - Slot utilization - consistently increasing or remaining high; for example, 90% or higher.
 - If slot utilization has spikes, are they on regular frequency?
 - Some of non-critical workload can be time shifted
 - For given set of priority/label/project - group of queries/users
 - Avg. wait time - consistently increasing
 - Avg. query runtime - consistently increasing





Pricing Models and Estimates

BigQuery pricing is separated by storage and compute



After your data is loaded into BigQuery, you are charged for storing it

Active storage pricing

Storage pricing is prorated per MB, per second. For example, if you store:

- 100 MB for half a month, you pay \$0.001 (a tenth of a cent)
- 500 GB for half a month, you pay \$5 (\$0.02/GB per month)
- 1 TB for a full month, you pay \$20

Long-term storage pricing

- Table or partition not edited 90+ consecutive days
- Pricing drops $\approx 50\%$
- No degradation (performance, durability, availability, functionality)
- Applies to BigQuery storage only

Different actions reset the timer

Activities that reset the timer

- Loading, copying, or streaming data into a partition or non-partitioned table
- Writing query results to a partition or non-partitioned table
- Using Data Definition Language

Activities that don't reset the timer

- Querying a table
- Creating a view that queries a table
- Exporting data from a table
- Copying a table to another destination table
- Patching or updating a table resource

Which BigQuery pricing model to pick?

On-demand pricing

- \$5/TB of data processed
- Quota limit of 2,000 slots
- Slots shared among all on-demand users
- 1st TB of data processed is free each month

Good for **exploratory work** and discovery

Flat-rate pricing

- Fixed rate pricing is \$10k* per 500 slots per month
- Slots are dedicated 24/7
- Starting at 100 slots
- Unlimited use of slots

Good for **multiple large workloads** and **consistent monthly billing**

BigQuery slots: Predictable and flexible pricing

Mix and match pricing models to get the best value for money

1

Pay per query

Bytes processed; pay as you go

2

Flex slots

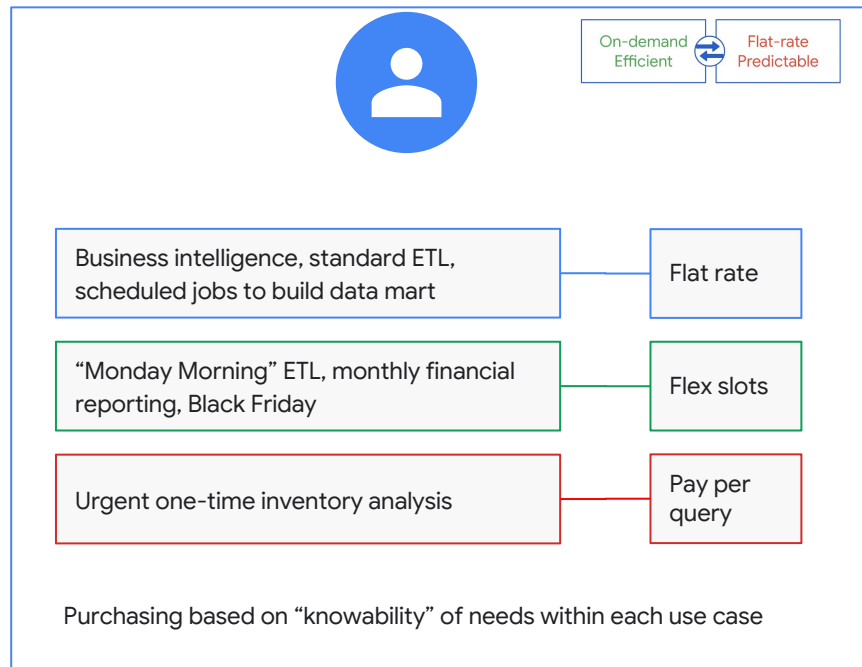
Burst capacity; pay as you go; Minutes/Hours/Days

3

Flat rate

Predictable lowest price; Monthly/Yearly

* Storage, Streaming, and BI Engine are charged separately



* Ref: [BigQuery Analysis Pricing Model](#)

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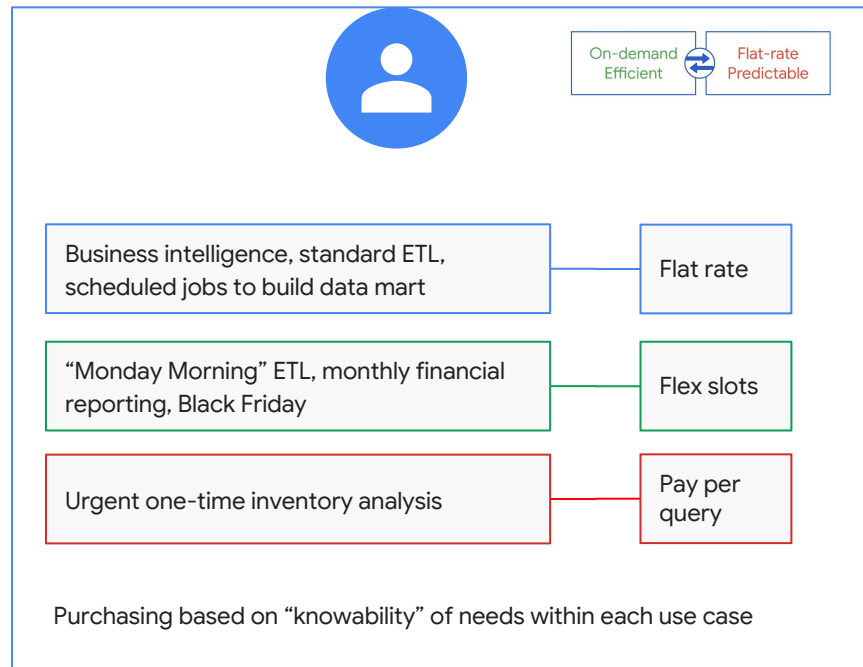
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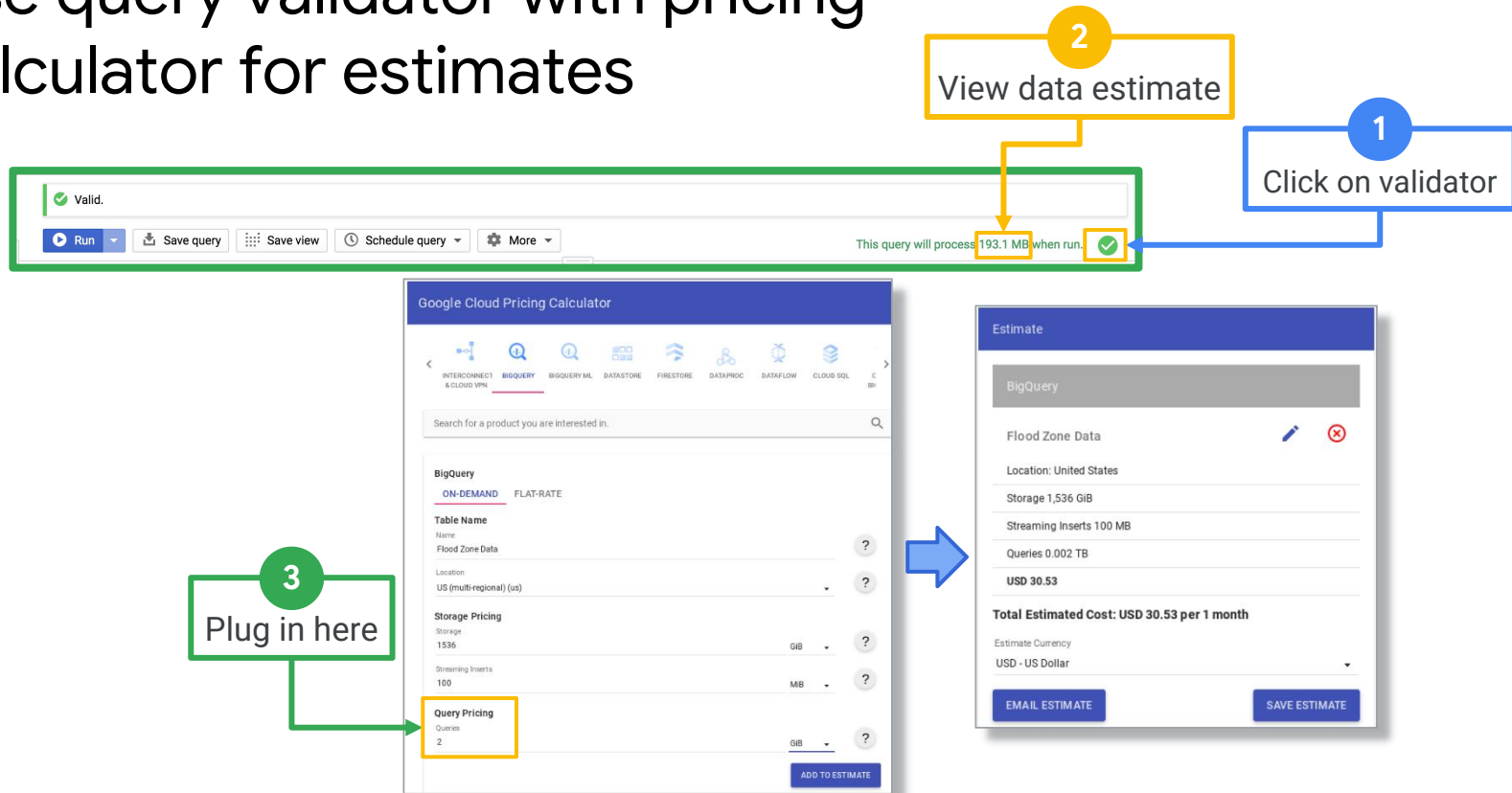
Predictable lowest price; Monthly/Yearly

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Use query validator with pricing calculator for estimates



Estimating query costs: Determine the query size through the command line

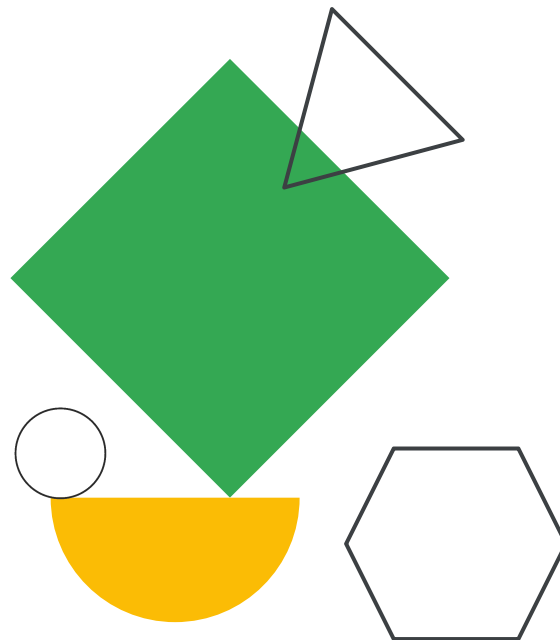
```
bq --location=[LOCATION] query --use_legacy_sql=false --dry_run  
'SELECT [FIELD1], [FIELD2], [FIELD3] FROM  
`[PROJECT].[DATASET].[TABLE]` LIMIT 1000'
```

Query successfully validated. Assuming the tables are not modified, running this query will process 10918 bytes of data.

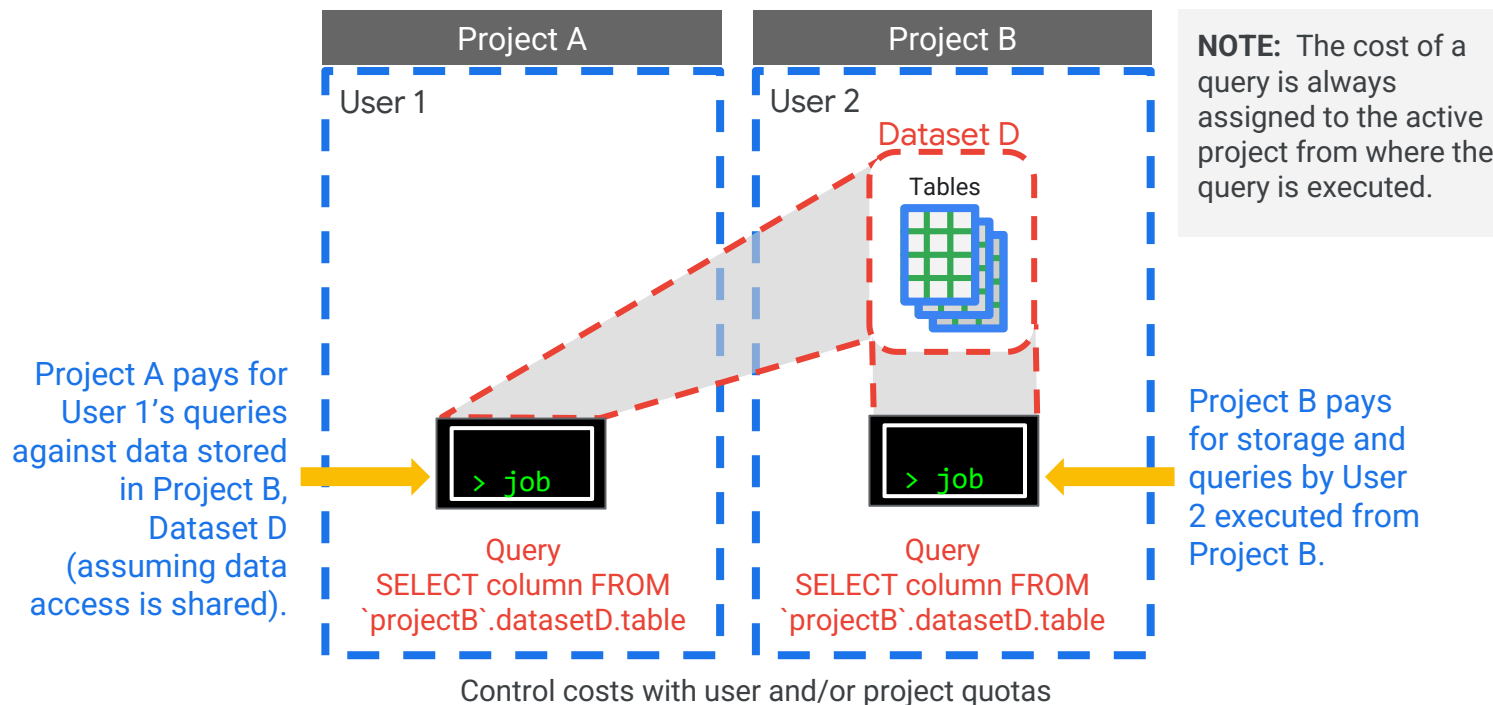
Demo

Determining pricing in BigQuery

Using the pricing calculator, bq
dry_run, and the API



You can separate cost of storage and cost of queries



For queries, on-demand pricing is flexible and efficient



On-demand pricing is based solely on usage.



It doesn't matter where the data is stored.



You can set user-level and project-level custom cost controls.



You can set the maximum bytes billed by query.

Several things to note regarding query charges

- Charged according to the total data processed in the columns selected and type of data.
- No charge for errors or results from the cache.
- Rounded to nearest MB; min 10 MB.
- Canceled queries may incur charges.
- Data processed in columns selected supersedes LIMIT set.
- Use partitioning and clustering to reduce data processed.
- On-demand pricing = analysis pricing.

BigQuery slots; Predictable and flexible pricing

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Flex slots

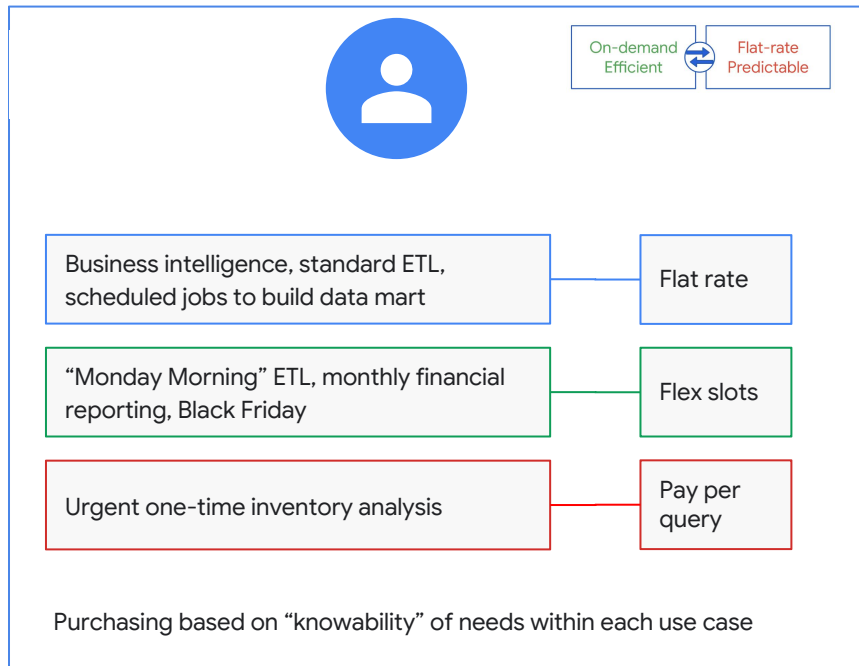
Burst capacity; pay as you go; Minutes/Hours/Days

3

Flat rate

Predictable lowest price; Monthly/Yearly

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Flex slots provide flexibility and control

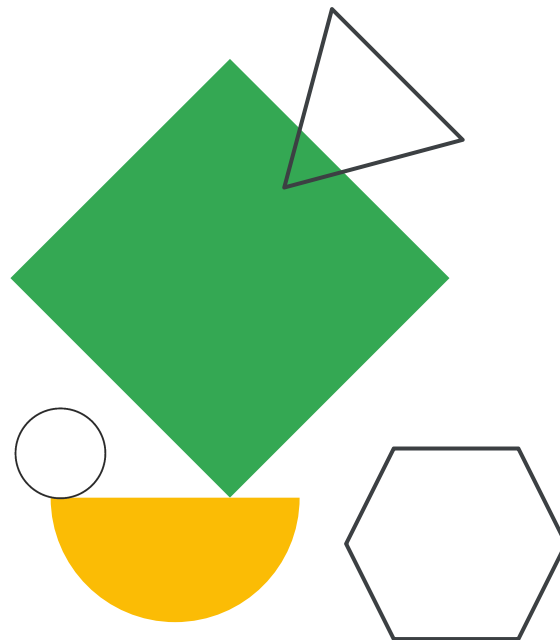
- ✓ Deploy BigQuery Slots in minutes.
- ✓ Only pay for what you consume.
- ✓ Cancel any time after 60 seconds.
- ✓ Combine with longer-term commitments.

Flex slots enable you to scale

- Planning for major calendar events, such as the tax season, Black Friday, popular media events, and video game launches
- Meeting cyclical periods of high demand for analytics, like Monday mornings
- Completing data warehouse evaluations and dialing in the optimal number of slots to use

Demo

Flex slots in BigQuery using
Reservations DDL statements





03

Slot Reservations

BigQuery slots; Predictable and flexible pricing

Mix and match pricing models to get the best value for money

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Flex slots

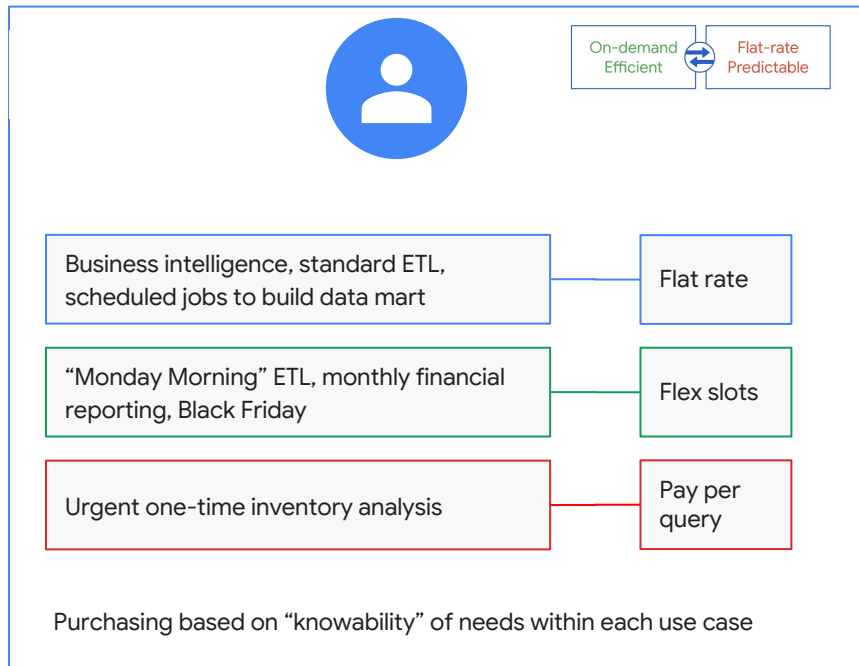
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Flat rate

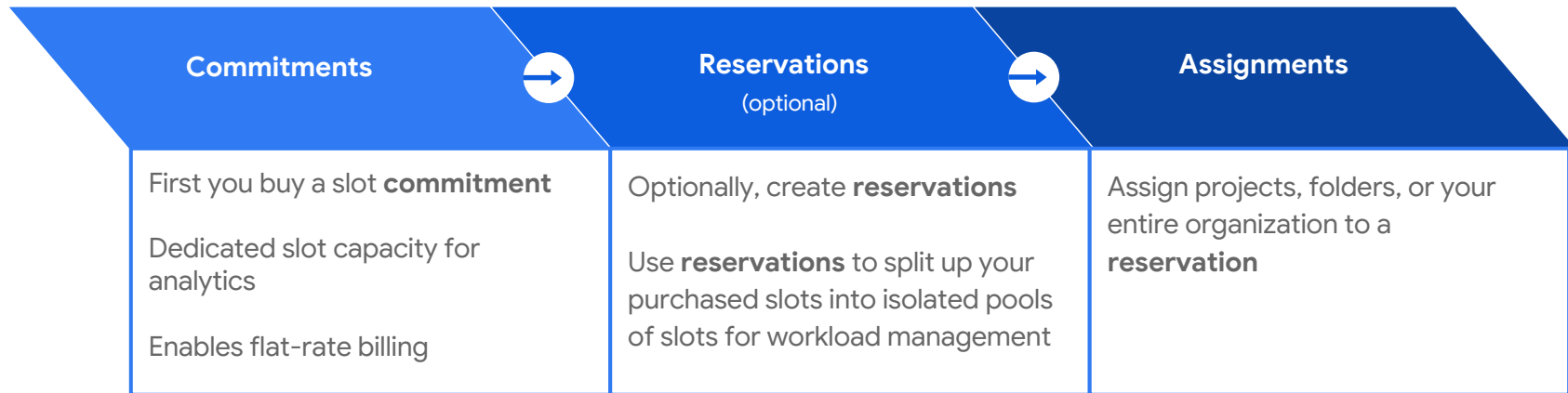
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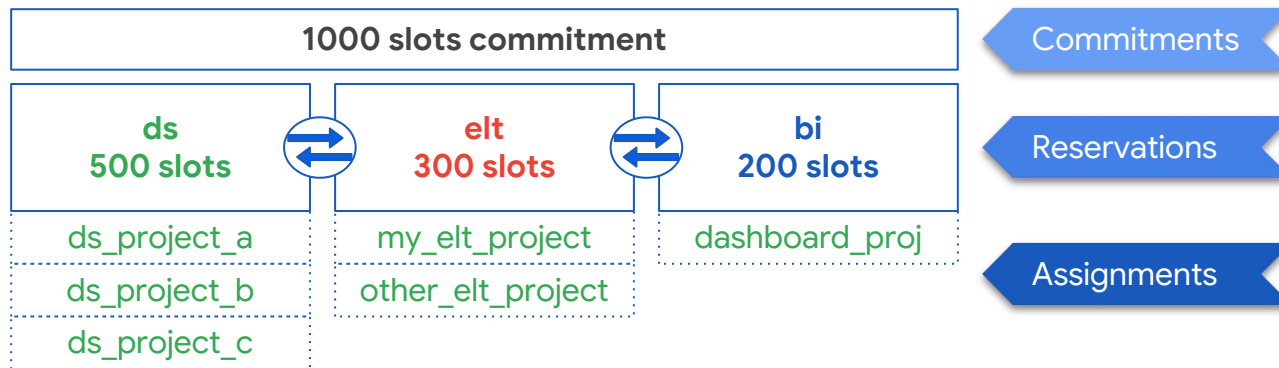
BigQuery slots: Flat rate terminology



- The first time you buy a slot **commitment**, BigQuery creates a default **reservation** and **assigns** the org to it.
- Admin project centralizes the billing and management of your commitments.

Slots flow

- Purchase slots at organization level (commitment).
- Create reservation and allocate slots to reservations per team or use case (reservation).
- Assign projects to respective reservations (assignment).
- Any unused capacity is seamlessly shared at org level ([idle slot sharing](#)).



Capacity commitment commitments

BigQuery Reservations [+ BUY SLOTS](#) [CREATE RESERVATION](#)

Capacity summary

Total slots: 1000

Unallocated slots: 2

[SLOT COMMITMENTS](#) [RESERVATIONS](#) [ASSIGNMENTS](#)

Commitments [SPLIT](#) [MERGE](#)

Enter property name or value

Status	Slots	Plan	Renewal plan	Commitment end time	Location	Slot commitment ID	Actions
☒	1000	MONTHLY	-	October 25, 2020 at 2:28:49 PM UTC-7	United States (US)	4158561784369798936	⋮

- Three different commitment plans: **annual, monthly, and flex**
- Must be purchased in **100-slot increments**
- Regional resources

Reservations and assignments

SLOT COMMITMENTS	?	RESERVATIONS	?	ASSIGNMENTS	?
Enter property name or value					
Name ↑	Slots	Location	Actions		
dannysandbox	250	United States (US)	⋮		
default	100	United States (US)	⋮		
load-default	100	United States (US)	⋮		
mixed-reservation	100	United States (US)	⋮		

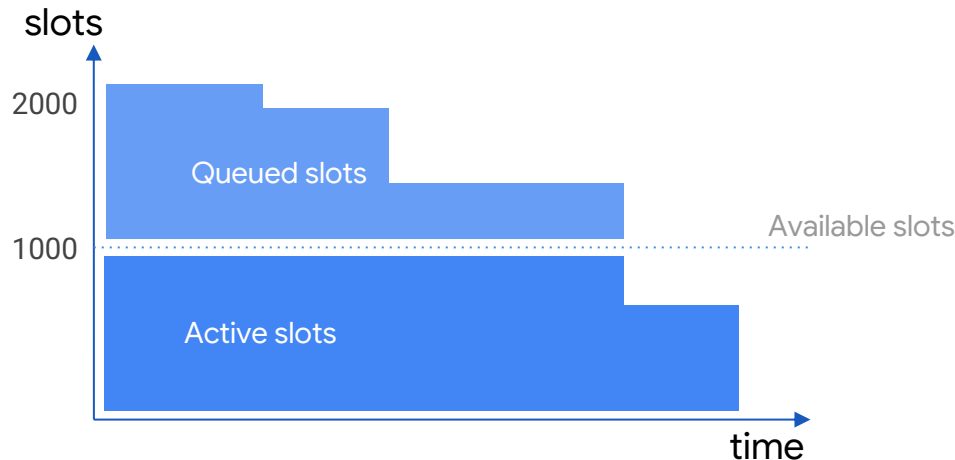
SLOT COMMITMENTS	?	RESERVATIONS	?	ASSIGNMENTS	?
Filter Enter property name or value					
Status	Name ↑	Reservation	Job Type		
✓	organizations/827442274178	projects/zr-prod-data-warehouse/locations/US/reservations/default	QUERY		
✓	organizations/827442274178	projects/zr-prod-data-warehouse/locations/US/reservations/load-default	PIPELINE		
✓	projects/jmeter-demo	projects/zr-prod-data-warehouse/locations/US/reservations/dannysandbox	QUERY		
✓	projects/tw-bq-storage-proj	projects/zr-prod-data-warehouse/locations/US/reservations/ml-dev	ML_EXTERNAL		
✓	projects/tw-dts-terraform	projects/zr-prod-data-warehouse/locations/US/reservations/mixed-reservation	PIPELINE		

- Reservation is an **isolated pool of slots** from a commitment.
- Assign **projects, folders, or organizations** to reservations to manage slot resources.
- Assignments for **QUERY, PIPELINE, or ML_EXTERNAL** types

Reservations: Resource queueing

Within a reservation

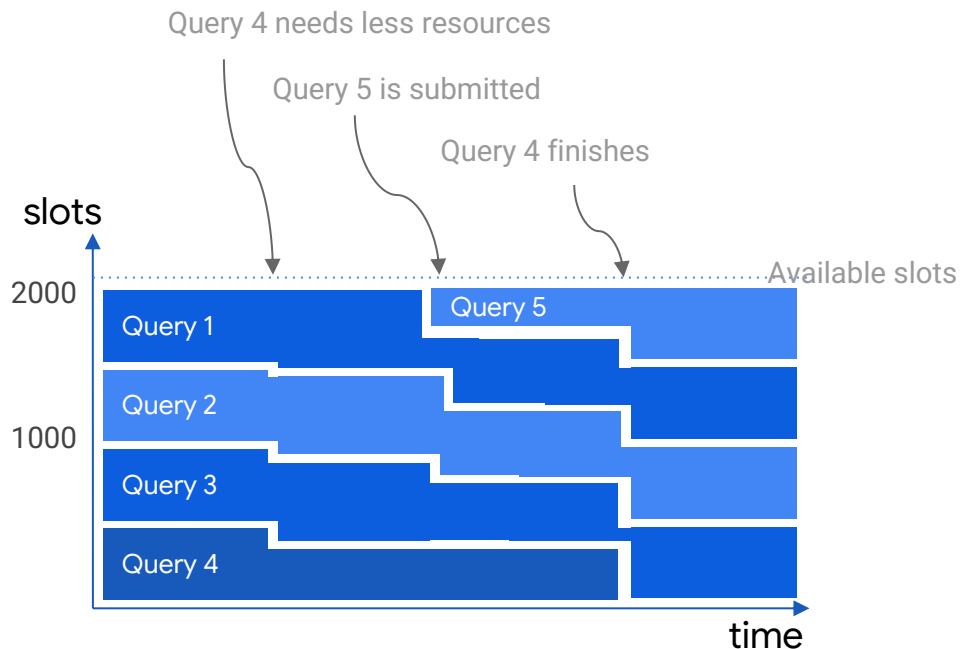
- If resource demand exceeds available capacity, BigQuery queues up additional slots.
- You are not charged for additional slots, and you are not charged for additional on-demand rates.
- As query execution progresses, BigQuery automatically works through the queued-up work until none is left.



Reservations: Fair sharing

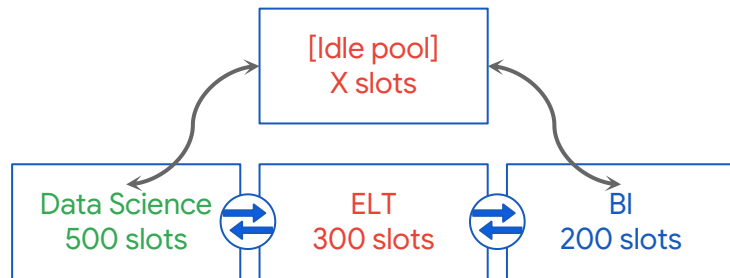
Seamless resource reallocation for queries

- BigQuery dynamically allocates capacity.
- Each query gets “fair share” of resources:
 - Slots are distributed fairly across all projects.
 - Slots are distributed fairly across all active queries within the projects.
- Subdividing of compute resources will continue to happen as more queries are executed.
- **Avoids workload starvation.**



Reservations: Idle slot sharing

- Any idle capacity in a reservation is seamlessly available for other reservations to use.
- As soon as the reservation needs that capacity back, it gets it.
- Queries whose work was preempted will not fail: they simply go back to using their resources as before.
- This happens in real time for every slot.
- The net effect is that all capacity in an organization is available to be used at any time.



Flat-rate pricing offers predictable and consistent month-to-month costs



Applies to query costs, including DML and DDL statements.



Doesn't apply to storage costs.



Purchased as a regional resource.



Available in monthly and annual commitments.



Customers can raise per-project concurrency quotas by contacting Google Cloud Support.

Reservations: Unusual applications

Things we've observed our customers doing that we've never planned for:

- Zero slot reservations for running jobs with background priority. The system will only ever run these jobs with idle slots when available.
- Creating penalty box reservations with a small number of slots, with idle slots usage disabled. Used for penalizing overly expensive queries.
- Reservations called high-pri/low-pri with frequent movement of jobs in and out of these reservations. Note that if high-pri reservation is unused, its idle slots can still be used by the low-pri reservation.

(*) any unused capacity seamlessly shared

Considerations when enrolling for flat-rate pricing

- Flex slots are a special commitment type and are subject to capacity availability.
- Monthly commitments cannot be canceled or downgraded for 30 calendar days from the purchase confirmation date.
- Annual commitments cannot be canceled or downgraded for one calendar year.
- To purchase additional BigQuery slots, you must enter into a new commitment.
- Flat-rate pricing is purchased for a specific BigQuery location.
- Flat-rate and on-demand pricing can be used together.
- To discontinue a flat-rate pricing plan, you must cancel or downgrade your commitment.



Controlling Costs

Control costs (1/5)

Avoid `SELECT *`

Query only the columns that you need.

Using `SELECT *` is the most expensive way to query data.

Sample data using preview options

Use table preview options to view data for free and without affecting quotas.

Control costs (2/5)

Price your queries before running them

Before running queries, preview them to estimate costs.

Queries are billed according to the number of bytes read.

Limit query costs by restricting the number of bytes billed

Use the maximum bytes billed setting to limit query costs.

Error: Query exceeded limit for bytes billed: 1000000. 10485760 or higher required.

Control costs (3/5)

LIMIT doesn't affect cost

Do not use a LIMIT clause as a method of cost control.

Applying a LIMIT clause to a query doesn't affect the amount of data that's read, unless clustering is used.

View costs using a dashboard and query your audit logs

Create a dashboard to view your billing data so you can make adjustments to your BigQuery usage.

Control costs (4/5)

Use partitioned and/or clustered tables

Partitioning and/or clustering your tables may improve performance by reading less data for certain queries, thereby reducing costs.

Materialize query results in stages

If possible, materialize your query results in stages.

Control costs (5/5)

Consider the cost of large result sets

Use the default table expiration time to remove the data when it's no longer needed.

Use streaming inserts with caution

There's no charge for loading data into BigQuery.

There is a charge, however, for streaming data into BigQuery.

Questions?

